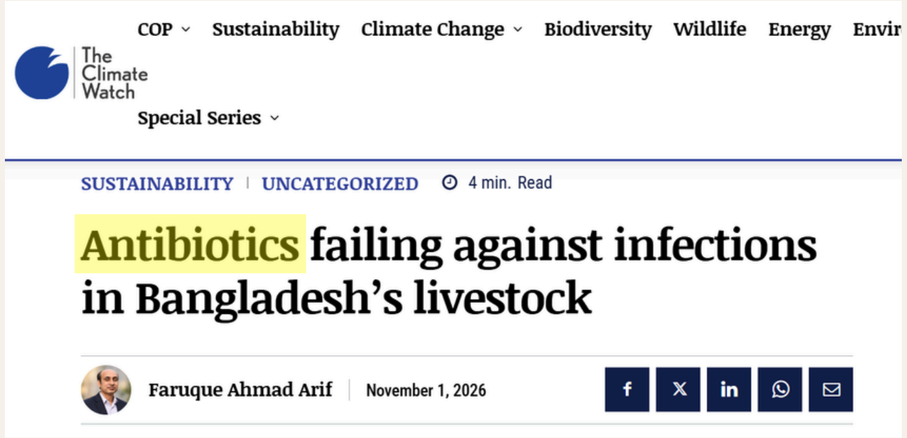
INSTITUT Pasteur

What’s on a bacteria’s mind ?

Understanding the bacteria’s decision making process

The Stakes — Antibiotic Resistance



Institut Pasteur & Microbiology : a love story

"The role of the infinitely small
is infinitely great"

Louis Pasteur

10 Nobel Prizes
in medicine (since 1907)

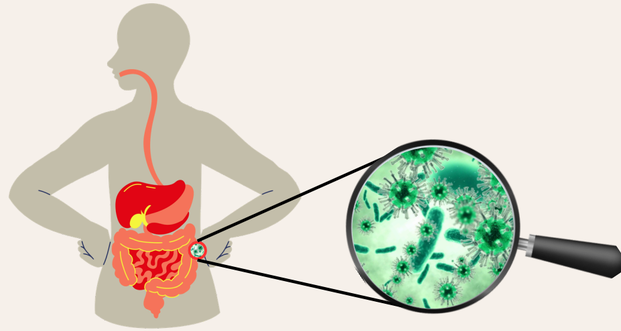
3,213
staff members on the Paris campus

93
different nationalities

32
Pasteur Network member institutes

Dead organic matter

Gut microbiota

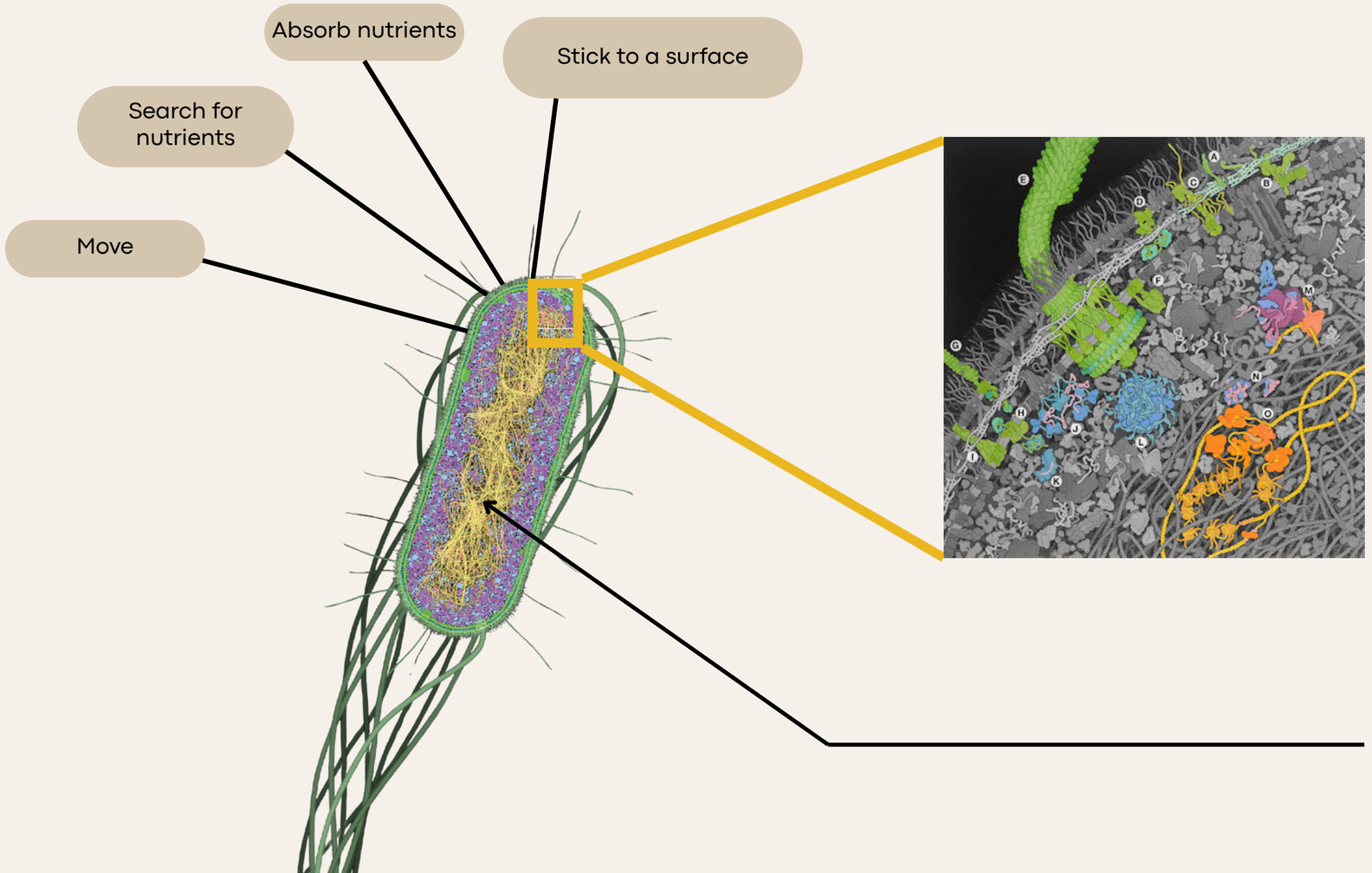


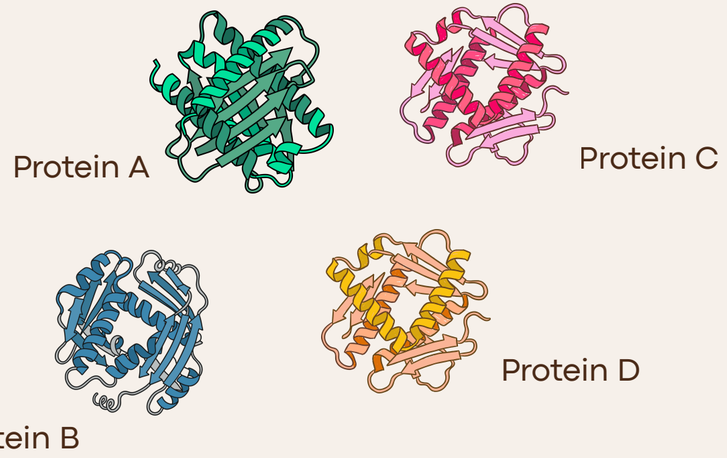
Fermentation

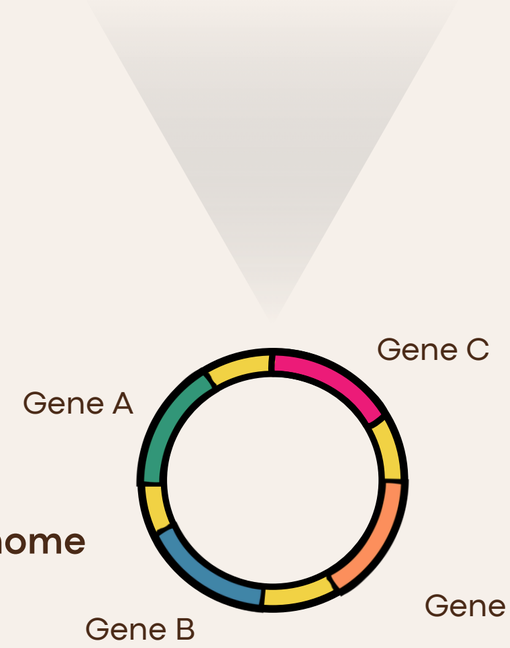
Cleaning the ocean
Producing 0_2

→ Global Leader in microbiology

The bacterial machinery







The Core Mystery — Adapting to the Environment

Lack of food → Explore

Environment

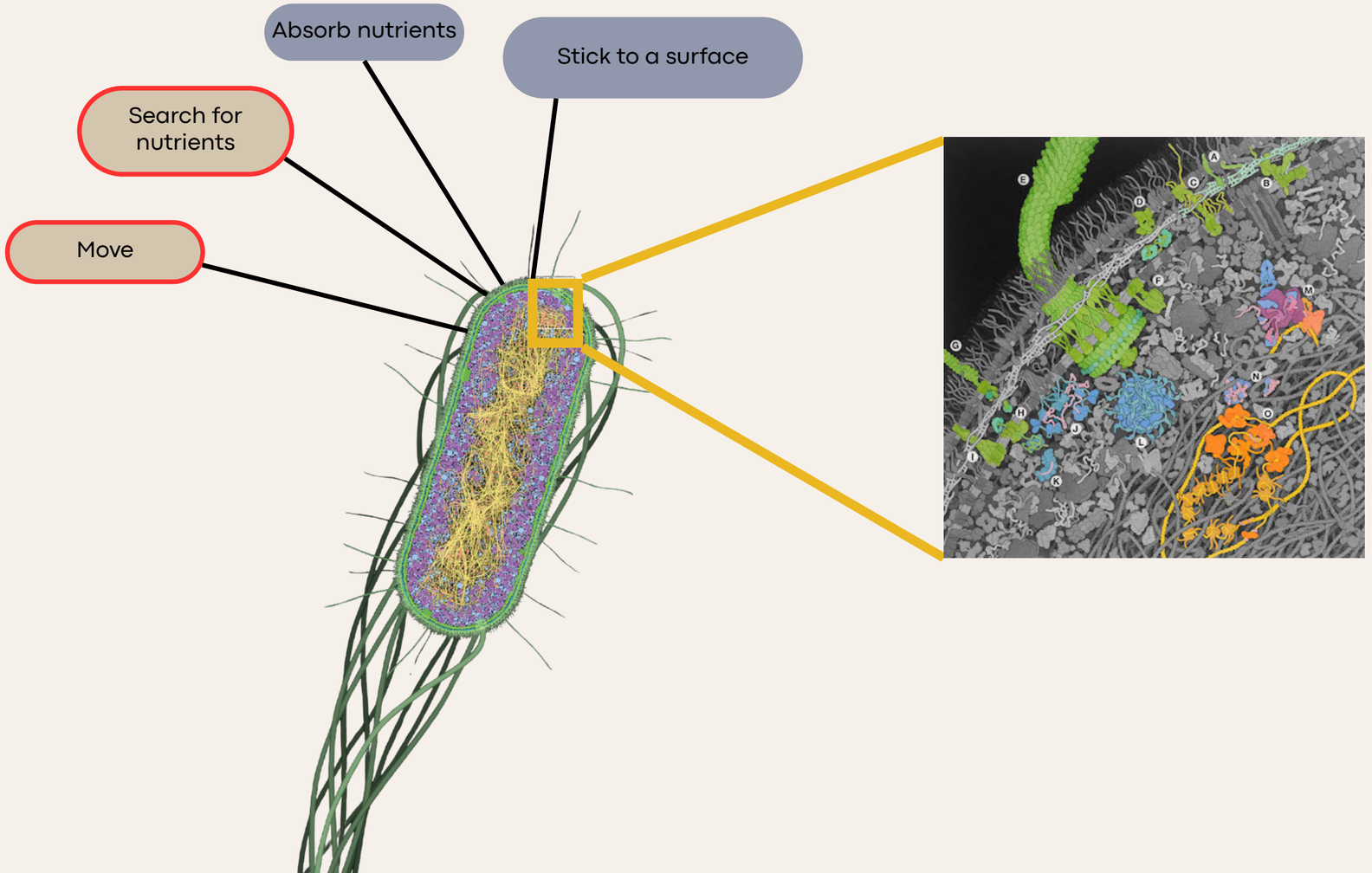
Reaction

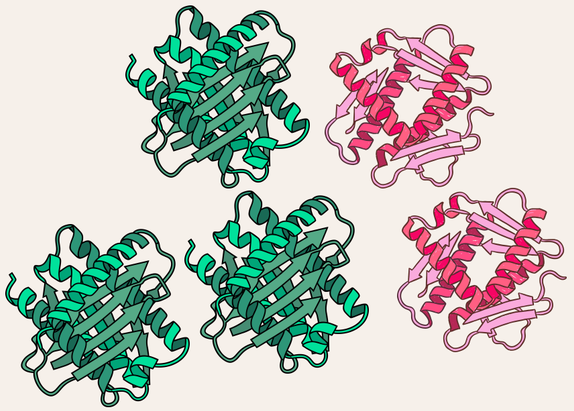
Absorb nutrients

Stick to a surface

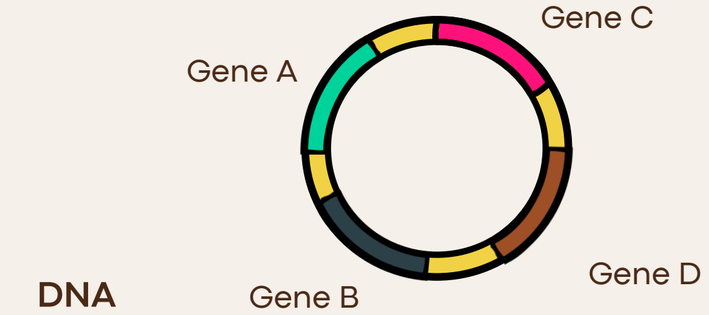
Search for nutrients

Move



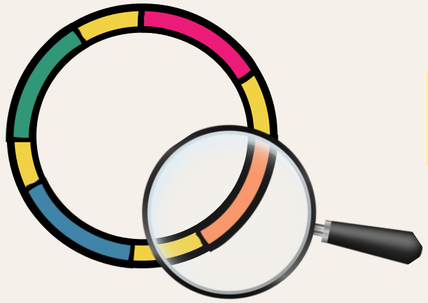


?



DNA

How to decrypt DNA



ATCGGCTAAGCTAGCGCATCGGATCGCGGATCGAT...CTAAGCTAGCGCATCGGCTAAGCTAGCGCATCGGATCGCGGAT

production
if food available

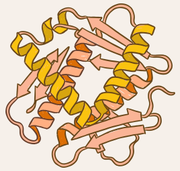
Promoter

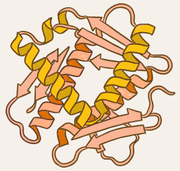
Gene D

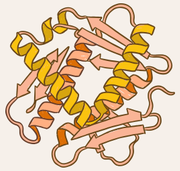
Gene F

ATCGGCTAAGCTAGCGCATCGGATCGCGGATCGAT...CTAAGCTAGCGCATCGGCTAAGCTAGCGCATCGGATCGCGGAT

production
if no food

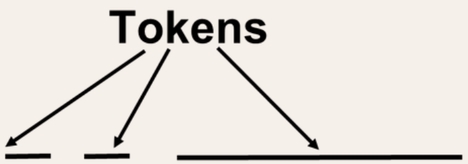




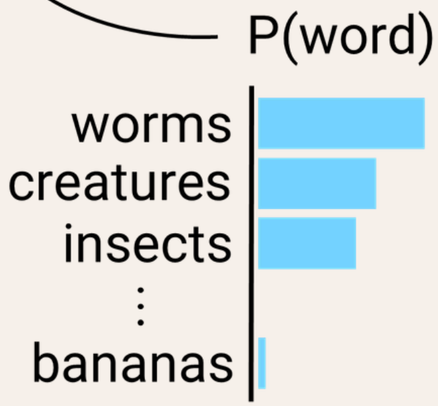


Biological Language Models

Natural language model (BERT)

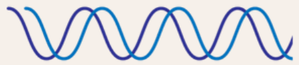


It is interesting to contemplate an entangled [redacted], clothed with many plants of many kinds, with [redacted] singing on the bushes, with various insects flitting about, and with [redacted] crawling through the damp earth ?



Brixi *et al. Nature*, 2026

DNA language model (EVO)



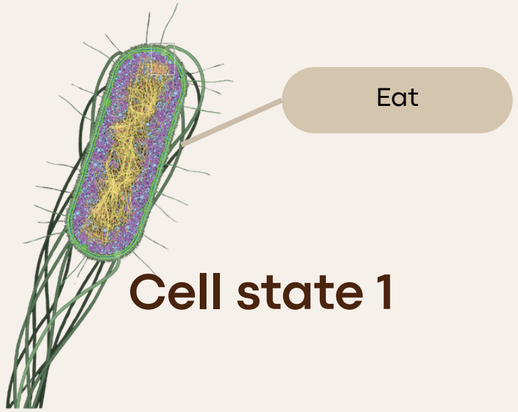
```
ACACACAGGACTG ACGAAT TTGA GC AATATGG CA
A GAAGC  A  TACGGACG TGGTTGATAGC C C TCA
C  G  AT CG  TTCTCT TG TGTCG GATGA CCACAC
CGG  TGCCACGCT GCG A AT  AGG G A CAGT  G
GCG GTT TATCTT  TGG GACGAAG GAAA AATCGCTG
GTGAA GC
```

Lin *et al. Science*, 2023

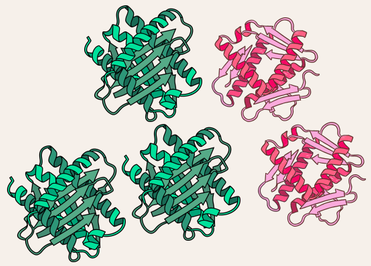
Protein language model (ESM)

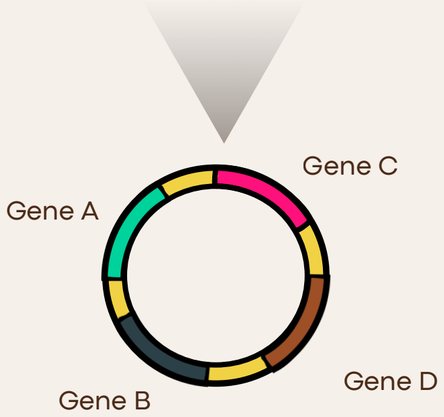
```
VVGKGQAANPVAC HCNNAP VTACPVN LTFQSDSVQL
D QKCIG  R  IACPFGVV MVD IAQKCDL N R SGTQ
A  E  PT AL  MDDKGL QI VARQR TAAGK SSDAQP
SRS  LPVNSRKGA KIS S RK  FGE Y GLDPQQ  Y
ESDRCVY AE ANC WHCPLHNA PDY RLVQEG I EAAE
 HQT  L
```

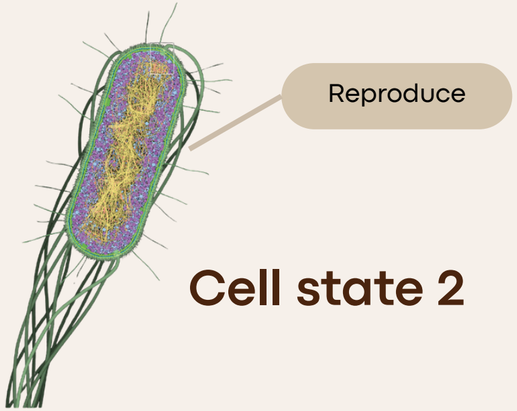
DNA → protein : Cell state



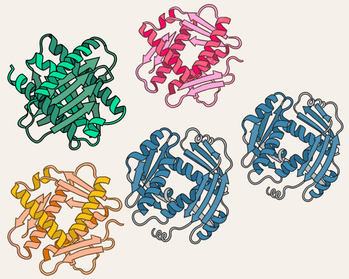
Cell state 1

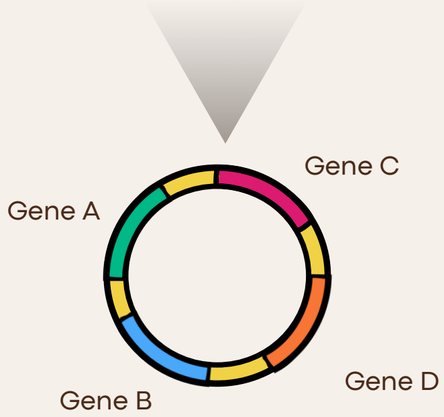


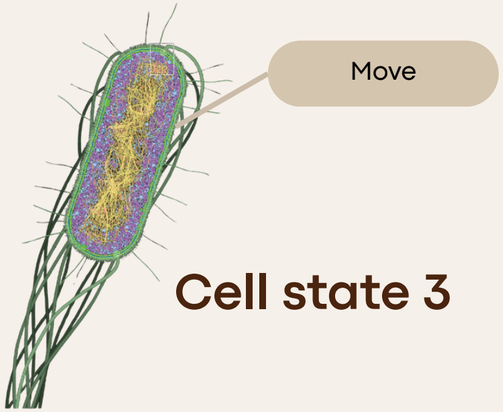




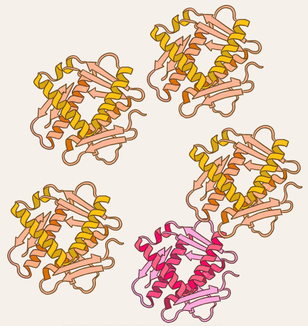
Cell state 2

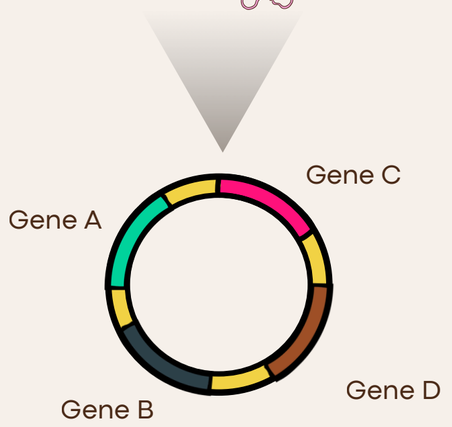




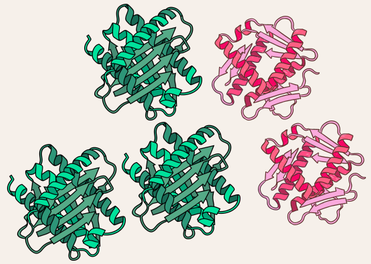


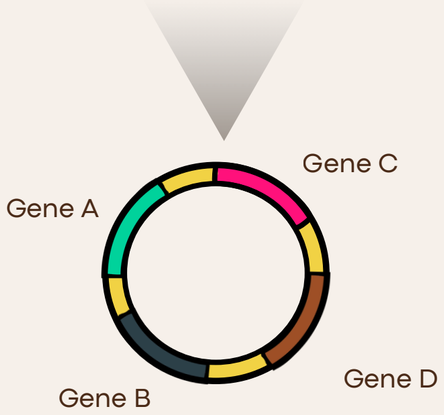
Cell state 3

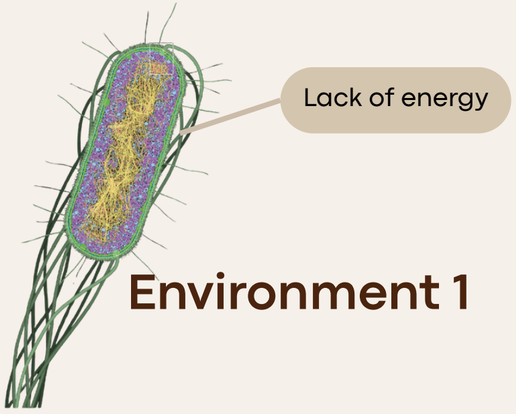




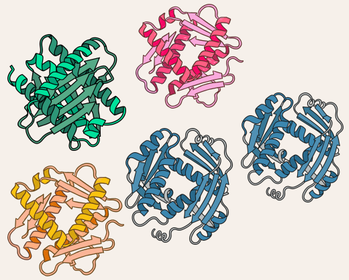
The Missing Variable — The Environment

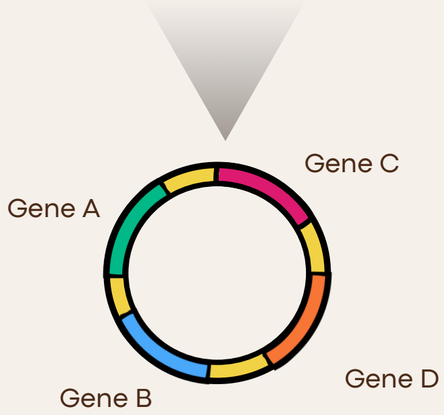


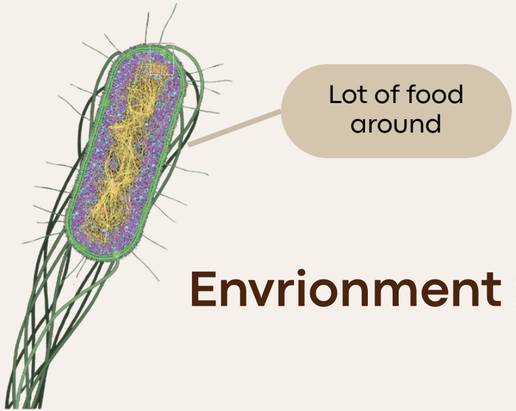




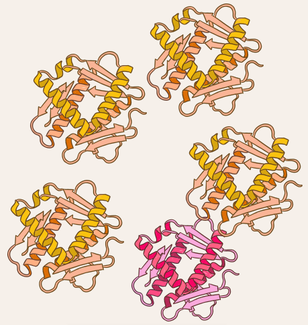
Environment 1

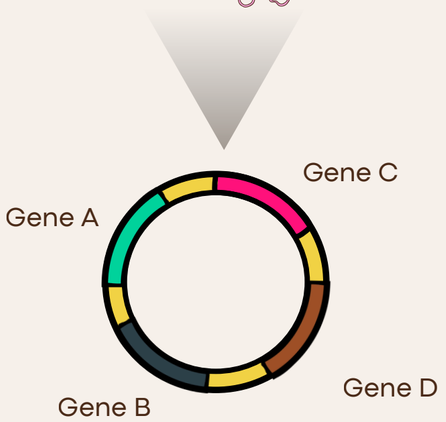


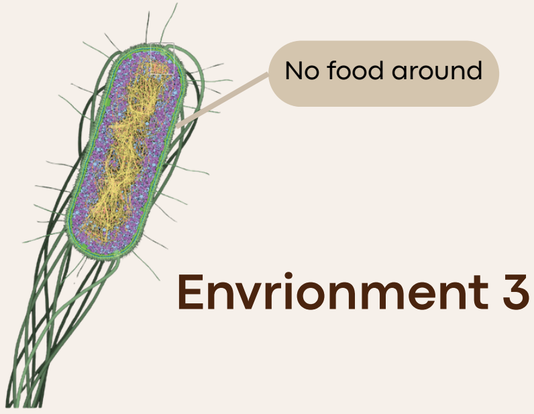




Envrionment 2

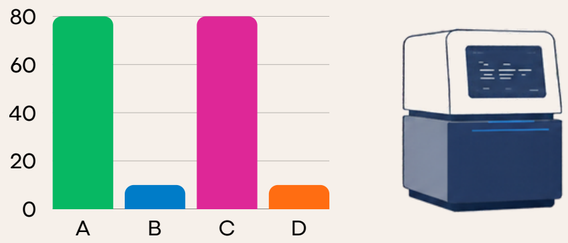


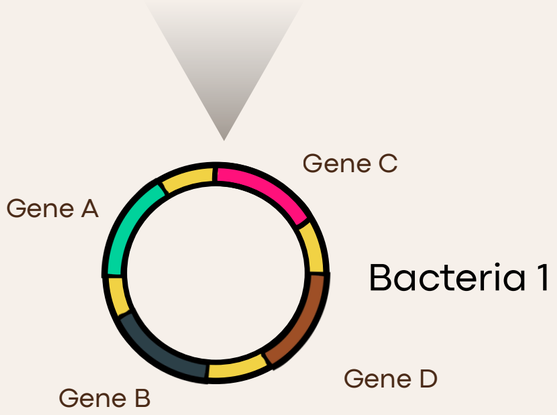


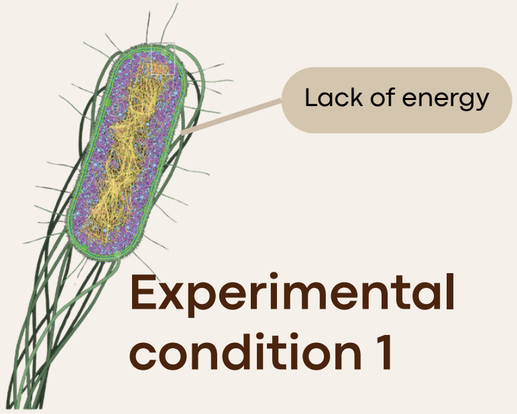


Envrionment 3

How We Measure This in the Lab (Read Counts)

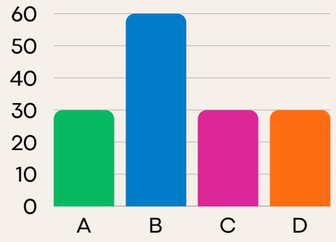


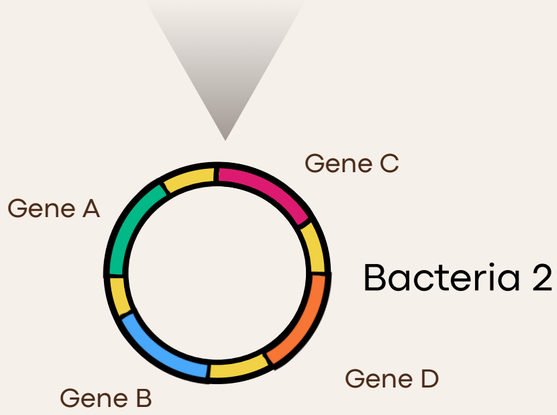


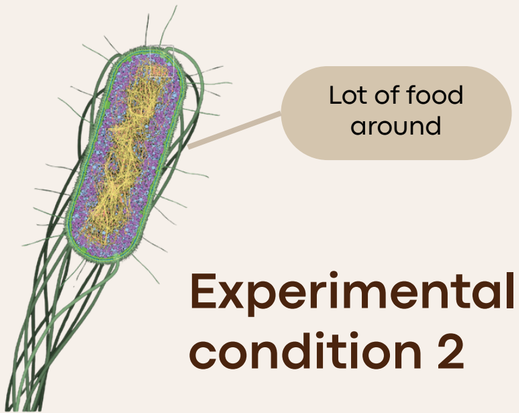


Lack of energy

Experimental condition 1

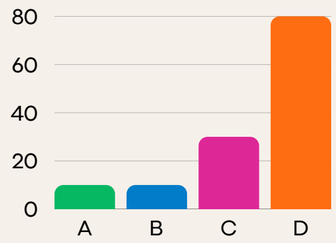


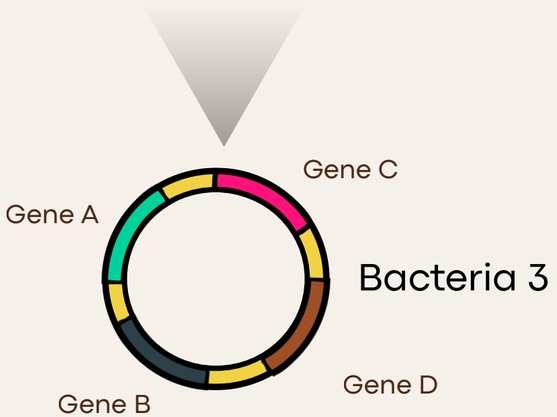


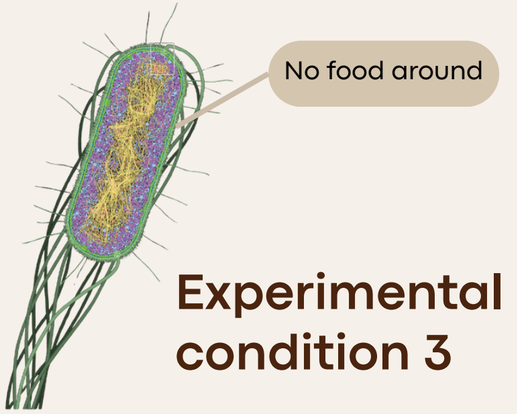


Lot of food around

Experimental condition 2



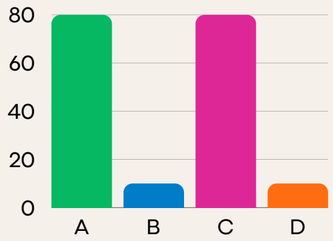


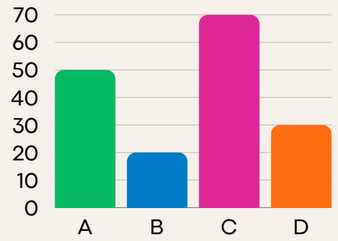


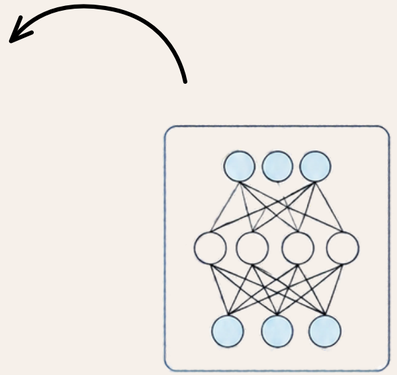
No food around

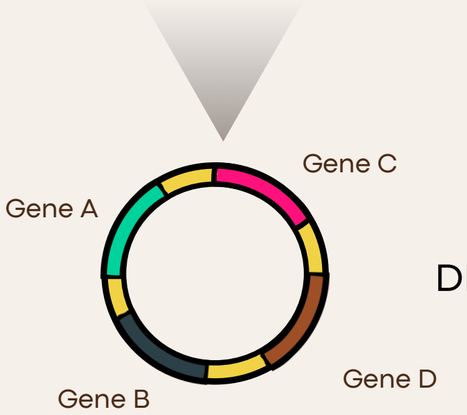
Experimental condition 3

Our Model — From

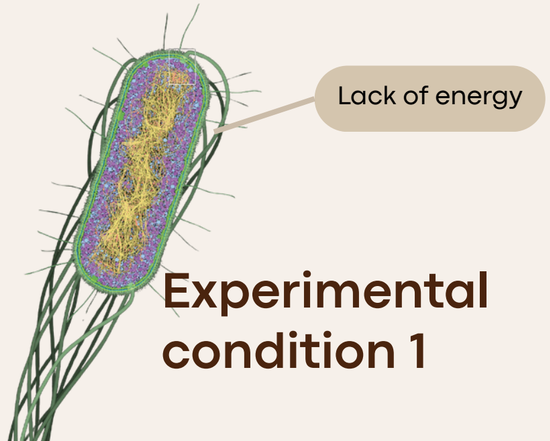








DNA



Experimental condition 1

What’s the challenge about ?

~19,500
RNA-seq samples

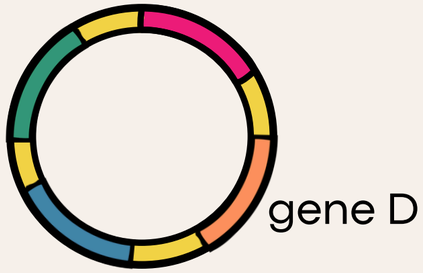
~9,000
experimental conditions

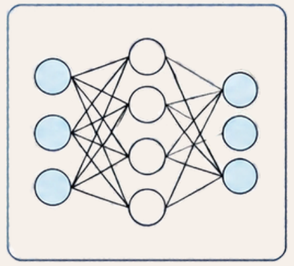
88
bacterial species

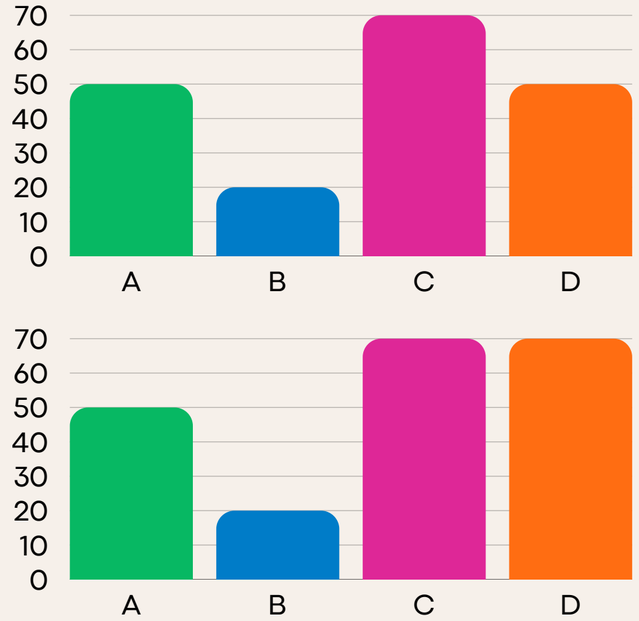
243
genomes

Experimental conditions + Specific Genome → Your model → Read counts (gene expression)

Temp °C	38
pH	6.7
Oxygen	aerobic
Growth	exp
Carbon source	glycerol



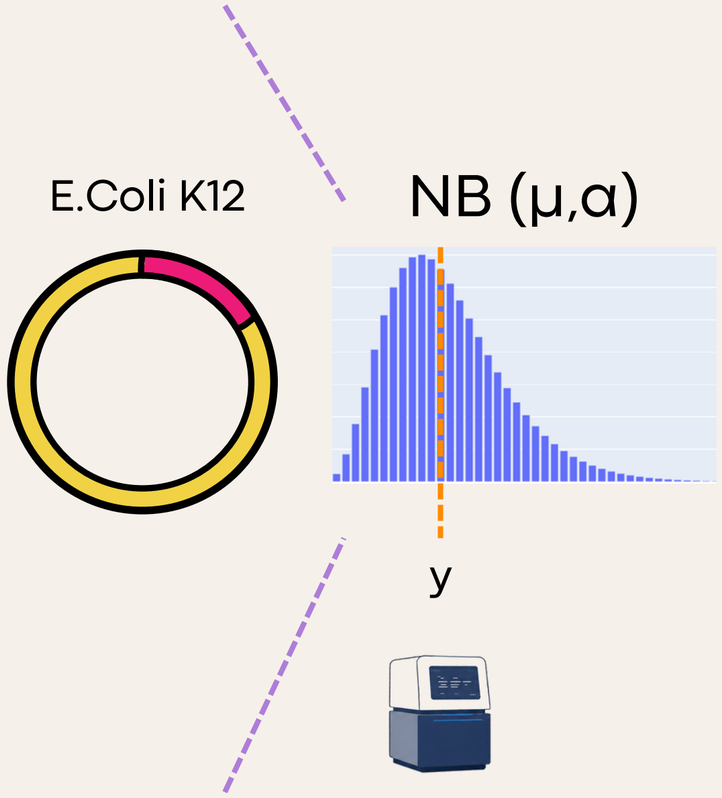




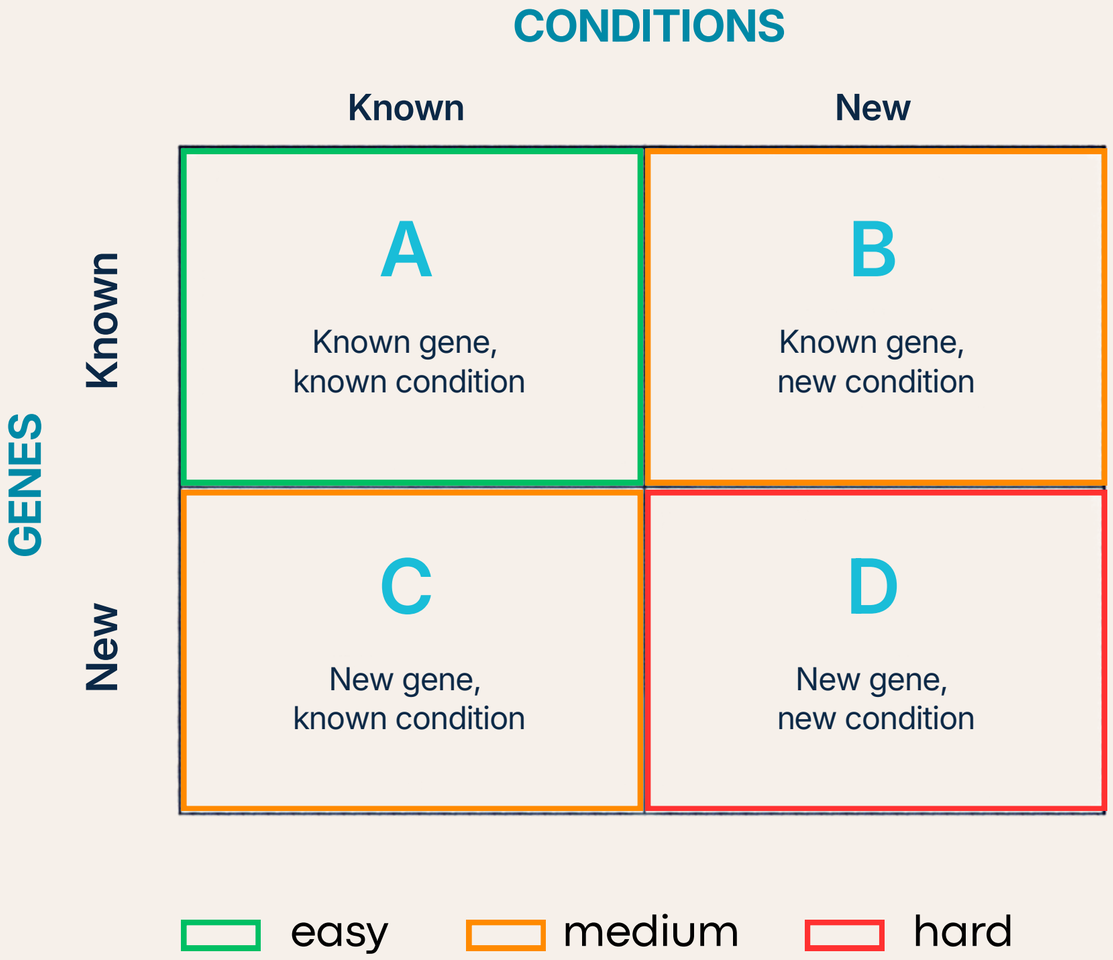
Implementation details : 1 instance

condition + gene → read count

Temp °C	**38**
pH	6.7
Oxygen	aerobic
Growth	exp
Carbon source	glycerol
Genetic modification	deletion
Target gene	hp_098



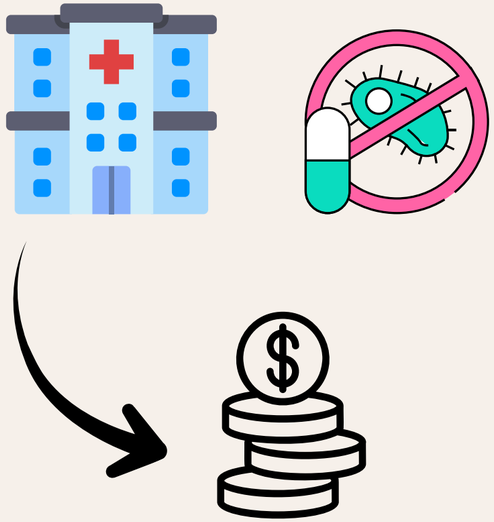
Test split



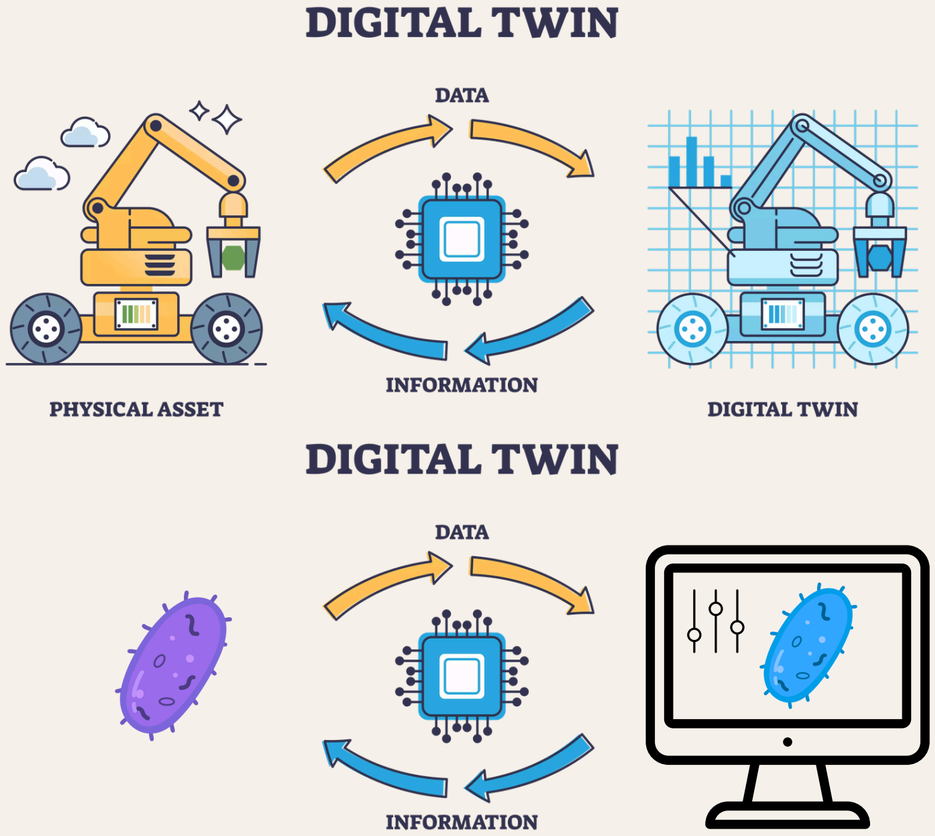
Perspective of this project

- Short term : paper publication on a groundbreaking research project

- Middle term : solve antibiotic resistance

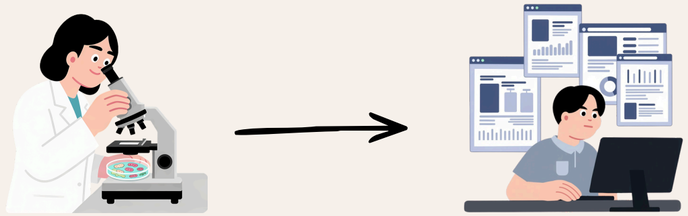


- Long term : simulate life



Skills that you can learn from this project

- Work with biological data



- Start working with gLMs : the new era of foundation models

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- Solving global health issues with AI: find purpose in your future work

Contributors

Ernest Mordret
FIG
Functional Interpretation of microbial Genomes

Lucas Paoli
Paoli Lab
Microbiome Immunity and Ecology